

RES-5 endpoint 2PI bound (v1.2, projection-corrected): the corrected

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

res5-tail-budget-closure localised RES-5 to the $I = 2 \times 10^{-3}$ endpoint. This note gives the corrected endpoint status. Two upstream corrections are now folded in: SC-SCOPE is LIFTED@THIN-CERTIFIED via the QUARTIC route (B1 is T6 on {H-LAYER} alone), and the projection closure lever is ELIMINATED ($\chi_{\text{proj}} = 1.25 > 1$, res5-projection-factor-bound), with the endpoint tail corrected upward from 0.047 to 0.059. The endpoint closes ONLY at the verified floor; its sole rigorous lever is the DR-2 floor κ . Scope: the $I = 2 \times 10^{-3}$ endpoint; estimate/verified grade as marked. No tier action.

2. The SC-SCOPE certified joint and its floor input

The endpoint slack $1 - 1/\text{joint}$ is set by the SC-SCOPE CERTIFIED joint (scscope-quartic-normalisation-certificate), with the floor ρ_{lat} as input:

$$\begin{aligned} \text{conservative } (K_{\text{floor}} \leq T') : \quad & \rho_{\text{lat}} = 6.55, \quad \text{joint} = 1.040, \quad \text{slack} = 0.0385, \\ \text{verified } (K_{\text{floor}} \leq 0.52 T' \text{ sample}) : \quad & \rho_{\text{lat}} = 12.6, \quad \text{joint} = 1.082, \quad \text{slack} = 0.0758. \end{aligned}$$

$K_{\text{floor}} \leq T'$ is PROVED; the factor below 1 is realized (Section 4).

3. The corrected endpoint tail (projection route eliminated)

The pattern-dependent tail is $C_{\text{higher}}/\Delta F_{\text{margin}} = f_{\text{avg}} a_0$, where $f(k) = 1/(1 + \lambda' \chi_0(k))$ is the screened response factor and f_{avg} is its w_t^2 -weighted average over the {110} modulation transfers. Direct computation (res5-projection-factor-bound) shows the bubble $\chi_0(k)$ is forward-peaked, so the screening is MAXIMAL at $k = 0$ ($f(0) = C_G = 0.49$) and WEAKER at the {110} transfers; hence

$$\chi_{\text{proj}} \equiv \frac{f_{\text{avg}}}{C_G} = \frac{0.613}{0.492} = 1.25 > 1, \quad \boxed{C_{\text{higher}} = f_{\text{avg}} a_0 = 0.059}$$

The earlier operator-norm value $C_G a_0 = 0.047$ was a forward-channel UNDER-estimate; 0.059 is the corrected tail. The projection lever $\chi_{\text{proj}} \leq 0.82$ is ELIMINATED (it is 1.25); there is NO screening-projection closure route.

4. The floor- κ lever (route A): the SOLE remaining lever, DR-2-adjacent

The only lever tightening the joint from conservative ($\kappa = 1$) toward verified is $\kappa = K_{\text{floor}}/T'$, $K_{\text{floor}} = \sum_{t \neq 0} w_t^2/I^2$. Two rigorous results:

- (a) PROVED refinement (Cauchy-Schwarz + $t = 0$ subtraction): $K_{\text{floor}} \leq T'(1 - \|A\|_4^4/I^2) \leq T'(1 - 1/n)$.
- (b) EXACT scan over crystallographic single shells ($n \leq 42$, uniform): worst $\kappa = 0.75$ (at $R = 4$, $n = 6$), correcting the incomplete-sample 0.52.

The refinement (a) is weak for dense patterns ($1 - 1/n \rightarrow 1$); the tight worst-case $\kappa < 1$ over the DENSE admissible class is the additive-energy / circle-incidence problem = the DR-2 frontier.

5. Verdict: verified-floor closure only; sole lever DR-2

$$\text{slack}_{\text{proved}}(0.0385) < \text{tail}(0.059) < \text{slack}_{\text{verified}}(0.0758)$$

The endpoint closes at STRONG EVIDENCE ONLY at the verified floor (tail/slack = 0.78, a 22% margin), and is OPEN at the conservative/proved floor ($0.059 > 0.0385$). There is exactly ONE rigorous lever – the DR-2 floor $\kappa < \kappa_{\text{needed}} \approx 0.8$ over the dense admissible class. The projection route is eliminated.

Unification. SC-SCOPE is lifted@thin-certified (quartic-pinned); it is NOT a B1 named hypothesis. The RES-5 endpoint’s rigorous closure is the DR-2 additive-energy bound – the SAME frontier as B1’s unrestricted-class question.

Tier. RES-5/GAP-2: ENDPOINT-LOCALISED $\rightarrow$ STRONG EVIDENCE (off-endpoint $\geq 27\times$; endpoint at the verified floor, 22%). No tier flip: B1-RH-ENUM stays T6 CONDITIONAL on {H-LAYER}. The rigorous T6 endpoint needs the DR-2 κ bound (the projection alternative is gone).

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “v1.1 still said $\chi_{\text{proj}} < 0.82$ in the body.” UPHELD against v1.1: that text is removed here; the canonical value is $\chi_{\text{proj}} = 1.25$ and the tail is 0.059. v1.1 is superseded.
- (β) “Does the corrected tail break the verified-floor closure?” DISMISSED: $0.059 < 0.0758$ still holds (22% margin); only the conservative floor and the projection lever fail. Off-endpoint closure is unaffected.
- (γ) “Then nothing is proved at the endpoint.” ACCEPTED: STRONG EVIDENCE only; the sole theorem-grade path is the DR-2 $\kappa < 0.8$ bound over the dense class, beyond the exact single-shell 0.75.

Quantitative sanity check (mandatory): *bracket* $0.0385 < 0.059 < 0.0758$; *verified margin* $1 - 0.059/0.0758 = 22\%$; *projection* $\chi_{\text{proj}} = 1.25 > 0.82$ (lever eliminated); *exact scan* worst $\kappa = 0.75 \leq 1 - 1/6 = 0.83$.

7. Result footer

Result ID: B1-RH-ENUM / RES-5 endpoint 2PI bound (v1.2, projection-corrected)

Precise statement: The corrected endpoint tail is $f_{\text{avg}} a_0 = 0.059$ (the operator-norm $C_G a_0 = 0.047$ was a forward-channel under-estimate; $\chi_{\text{proj}} = 1.25 > 1$, res5-projection-factor-bound). Bracketed: $\text{slack_proved}(0.0385) < 0.059 < \text{slack_verified}(0.0758)$ of the SC-SCOPE CERTIFIED joint. Closes at STRONG EVIDENCE ONLY at the verified floor (22% margin), OPEN at the conservative floor. The projection lever is ELIMINATED; the SOLE rigorous lever is the DR-2 floor $\kappa < \sim 0.8$ over the dense admissible class (exact single-shell worst 0.75; proved $K \leq T'(1 - \|A\|_4^4 / I^2)$).

Scope: $I = 2e-3$ endpoint; STRONG EVIDENCE / verified grade as marked.

Dependencies: res5-tail-budget-closure (localisation); scscope-quartic-normalisation-certificate (certified joint; SC-SCOPE lift); res5-projection-factor-bound ($\chi_{\text{proj}} = 1.25$; tail 0.059); res5-a0-skeleton-sensitivity (C_G, a_0). Supersedes v1.0, v1.1.

Evidence grade: ANALYTIC + EXECUTED (res5_endpoint_2pi_bound.py 5/5 + res5_projection_factor.py 5/5) + INHERITED (certified joint).

Reproduction command: `python codes/vacuum/res5_endpoint_2pi_bound.py && python codes/vacuum/res5_projection_factor.py`

Expected output: tail 0.059 in $[0.0385, 0.0758]$; tail/slack_verified 0.78; $\chi_{\text{proj}} 1.25$; exact worst κ 0.75; 5/5 + 5/5 PASS.

Falsification gate: A DR-2 bound proving $\kappa \geq 0.8$ worst-case (endpoint open at every floor grade); or the certified joint failing.

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5/GAP-2: ENDPOINT-LOCALISED $\rightarrow$ STRONG EVIDENCE; rigorous T6 endpoint pending the DR-2 κ bound (projection alternative eliminated).

No-overclaim statement: No theorem. SC-SCOPE lifted@thin-certified (not a hypothesis); projection lever eliminated; tail corrected to 0.059; verified-floor closure only (22%), conservative floor OPEN. Sole rigorous residual = DR-2. B1 unchanged (T6 on {H-LAYER}).

Next required action: res5-dr2-kappa-bound: a worst-case additive-energy bound $\kappa = K_{\text{floor}} / T' < \sim 0.8$ over the dense admissible class (closes the RES-5 endpoint AND the unrestricted-class B1).